

Relativistic Mechanics

Labix

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1 Special Relativity

1.1 Minkowski Spacetime

Einstein's two postulates.

Postulate: Homogeneity of space and time and isotropy.

From the above, conclude that spacetime is Minkowski.

Definition 1.1.1 (Minkowski Spacetime)

Define the Minkowski spacetime by the following data.

- A four dimensional affine space $(M, V, +)$.
- A non-degenerate symmetric bilinear form $\eta : V \times V \rightarrow \mathbb{R}$ of signature $(3, 1)$.

By Sylvester's law of inertia, we are guaranteed global Lorentz orthonormal bases $E = \{v_0, \dots, v_3\}$. These are the bases for which

$$[\eta]_{E,E} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Typically, we think of η has having the matrix

$$\begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where c is the speed of time. But the geometry does not differ other than by a scalar because it only depends on the signature.

We recall some facts about affine spaces that makes them a smooth manifold.

- For each $p \in M$, the map $\phi_p : M \rightarrow V$ given by $\phi_p(x) = x - p$ does not define a global chart on M . We need to choose a basis for V to get an isomorphism $V \cong \mathbb{R}^4$ for the global chart.
- For each $p \in M$, the map $\alpha_p : T_p M \rightarrow V$ given by $[\gamma] \mapsto \left. \frac{d}{dt} \right|_{t=0} (\gamma(t) - p)$ is a canonical isomorphism. This means that no connection is needed to compare the fibers of the tangent bundle. The structure of the affine space means inherently that M is flat.

We also recall some definitions of Lorentzian geometry. Let V be a vector space of dimension 4. Let η be a non-degenerate symmetric bilinear form on V of signature $(3, 1)$. Let $v = (t, v_1, v_2, v_3) \in \mathbb{R}^4$.

- We say that v is space-like if $|v|_\eta > 0$.
- We say that v is time-like if $|v|_\eta \in \mathbb{C} \setminus \mathbb{R}$.
- We say that v is light-like if $|v|_\eta = 0$.

Definition 1.1.2 (Time-Orientation)

Let $(M, V, +, \eta)$ be Minkowski spacetime. A time orientation of M is a choice of connected component of the set of time-like vectors $\{v \in V \mid |v|_\eta \in \mathbb{C} \setminus \mathbb{R}\}$.

Definition 1.1.3 (Future Pointing Vectors)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M . Let $v \in V$ be a time-like vector. We say that v is future pointing if v lies in the connected component defining the time orientation.

Definition 1.1.4 (The Minkowski Metric)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Define the Minkowski metric $\eta \in \mathcal{C}^\infty(T^*M \otimes T^*M)$ as

follows. For each $p \in M$, define

$$\eta_p(v, w) = \eta(\alpha_p(v), \alpha_p(w))$$

where α_p is the isomorphism $\alpha_p : V \xrightarrow{\cong} T_p M$.

The Lorentzian properties on V translate canonically to $T_p M$ via α_p . This gives M the structure of a pseudo-Riemannian manifold.

Proposition 1.1.5

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then (M, η) is a pseudo-Riemannian manifold with metric signature $(1, 3)$.

For a curve $\gamma : \mathbb{R} \rightarrow M$ in Minkowski space time, the vectors $\gamma'(t) \in T_p M$ is either space-like, time-like or light-like. Solving the equation $|x|_\eta = 0$ gives a 3-dimensional cone in $\mathbb{R}^4$.

- If $\gamma'(t)$ lies on the light cone, then $\gamma(t)$ is travelling at the speed of light.
- If $\gamma'(t)$ lies inside the light cone (time-like), then this is physically realizable as an object travelling through space time.
- If $\gamma'(t)$ lies outside the light cone (space-like), then it is physically impossible.

Definition 1.1.6 (Inertial Frame of Reference)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M . An inertial frame of reference consists of the following data.

- A choice of the origin $p \in M$.
- A Lorentz orthonormal basis $e_0, \dots, e_3$ of V such that e_0 is future pointing.

Let $E = \{e_0, \dots, e_3\}$. Let $\phi_E : V \xrightarrow{\cong} \mathbb{R}^4$ be the induced isomorphism. Then $(M, \phi_E \circ \phi_p)$ is a chart on M .

Definition 1.1.7 (The Poincare Group)

Let V be a vector space of dimension 4. Define the Poincare group to be

$$P = O(1, 3) \ltimes V$$

Proposition 1.1.8

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then there is an isomorphism

$$\text{Isom}(M, \eta) \cong O(1, 3) \ltimes V$$

Proposition 1.1.9

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then the Poincare group is the largest subgroup of $\text{Aff}(V)$ that maps inertial frames to inertial frames. Also it acts transitively on inertial frames.

Definition 1.1.10 (Translations, Boosts and Rotations)

Let $T \in O(1, 3) \ltimes \mathbb{R}^4$.

- We say that T is a boost if T is of the form

$$T(t, x) = \left(\frac{t - v \cdot x}{\sqrt{1 - v^2}}, \frac{\langle x, v \rangle}{|v|} - |v|t + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right) \right)$$

for some $v \in \mathbb{R}^3$ such that $|v| < 1$, and $x = \frac{\langle x, v \rangle}{|v|^2} v + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right)$ is the orthogonal decomposition.

- We say that T is a translation if T is of the form

$$T(t, x) = (t + t_0, x + x_0)$$

for some $x_0 \in \mathbb{R}^3$ and $t_0 \in \mathbb{R}$.

- We say that T is a rotation if T is of the form

$$T(t, x) = (t, Gx)$$

for some $G \in SO(3) \leq SO^+(1, 3)$.

Boosts are different in Minkowski spacetime because we assume $c = 1$ and no velocity can reach $c = 1$. So $|v| < 1$.

In the Galilean spacetime these three types decomposes the Galilean group. In Minkowski spacetime, boosts and rotations are given by the Cartan decomposition $SO^+(1, 3) \cong \{\text{Boosts}\} \times SO(3)$ as a smooth manifold (not as a group). Combining the translations give the restricted Poincare group, which is not the full Poincare group.

Definition 1.1.11 (The Restricted Poincare Group)

Define the restricted Poincare group to be

$$P^+ = \mathbb{R}^4 \rtimes SO^+(1, 3)$$

Proposition 1.1.12

There is an isomorphism

$$P \cong P^+ \rtimes_{\phi} (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$$

where $\phi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Aut}(P^+)$ is given by:

1.2 Relativistic Kinematics

Definition 1.2.1 (World Lines)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a curve. We say that r is a world line if $r'(t)$ is future pointing for all $t \in \mathbb{R}$.

Physically: time is coordinate dependent, proper time is the absolute notion that everyone agrees on.

Definition 1.2.2 (Proper Time)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Let $t_0 \in I$. Define the proper time of r to be

$$\tau(t) = \int_{t_0}^t \sqrt{-\eta(r'(s), r'(s))} ds$$

for $t \in I$ and $t > t_0$.

Heuristically, we have

$$d\tau = \sqrt{-\eta(r'(s), r'(s))} dt$$

The choice of coordinate frame changes how η and dt is presented, but $d\tau$ stays the same. If r is light-like instead, notice that $d\tau = 0$. Photons traveling in vacuum does not experience time in the same sense.

Time Dilation: Choose an inertial frame. Then r is parametrized as $r(t) = (ct, x(t), y(t), z(t))$. Let $v = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$. Then we have $d\tau = \sqrt{1 - |v|^2} dt$. So time experienced by an observer is proportional to proper time by a factor depending on the velocity of the observer.

The Twin Paradox: Among all world lines r passing through p and q , the geodesic one (the straight line) has the largest proper time. The twin paradox says that the twin taking the bendy road with acceleration ages faster than the other twin whose trajectory is straight. Mathematically, $r(\tau)$ is the unit speed parameterization of r .

Definition 1.2.3 (Unit Parametrization)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Define the unit parametrization of r by

$$\hat{r}(s) = (r \circ \tau^{-1})(s)$$

Definition 1.2.4 (Lorentz Factor)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Define the Lorentz factor of r to be

$$\gamma = \frac{1}{\sqrt{-\eta(r', r')}}}$$

Definition 1.2.5 (Four-Velocity and Four-Acceleration)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line.

- Define the four-velocity of r by

$$u(s_0) = \left. \frac{d\hat{r}}{ds} \right|_{s=s_0} = \gamma r'(s_0)$$

- Define the four-acceleration of r by

$$a(s_0) = \left. \frac{du}{ds} \right|_{s=s_0}$$

Proposition 1.2.6

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Then the following are true.

- $\eta(u, u) = -1$.
- $\eta(u, a) = 0$.

In Galilean spacetime, the three dimensional slices of the world along time is orthogonal to the flow of time. In Minkowski spacetime, the same is true, but time flows differently according to each observer (inertial frames).

Definition 1.2.7 (Simultaneity Hypersurface)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $v \in V$ be a future pointing time-like vector. Let $p \in M$. Define the simultaneity hypersurface through p relative to v by the space

$$\Sigma_p(v) = \{q \in M \mid \eta(q - p, v) = 0\}$$

A key point is that $\eta|_{\Sigma_p(v)}$ is positive definite, which means that it becomes an inner product on the three dimensional slice. Another key point: for the worldline r_1, r_2 of two observers at the same point p , their future pointing time-like vectors r'_1 and r'_2 are generally different, which gives a different three dimensional slice.

Length contraction: Suppose an metal rod exists in spacetime. The endpoints are given by two world lines r_1, r_2 such that $r_2 = q + r_1$ for some $q \in V$ and $\eta(r_2 - r_1, v)$ where here we choose an inertial frame to fix a future pointing vector v . Then define the length of the rod according to this observer to be the quantity

$$L = \sqrt{\eta(q, q)}$$

Then???

If a basis E is chosen for V such that

$$[\eta]_{E,E} \begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then unit parameterization would instead be

$$\tau(t) = \frac{1}{c} \int_{t_0}^t \sqrt{-\eta(r'(s), r'(s))} ds$$

and we would have $\eta(u, u) = -c^2$. Time dilation is displayed as $d\tau = \sqrt{1 - |v|^2/c^2} dt$

1.3 Relativistic Lagrangian

Four Momentum Four Force

1.4 Relativistic Statics

2 Classical Field Theory

Definition 2.0.1 (Classical Fields)

A classical field is an element

$$s \in \mathcal{C}^\infty(E)$$

where $E \rightarrow M$ is a vector bundle.

2.1 Lagrangian Density

Definition 2.1.1 (Lagrangian Density)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. A Lagrangian density on E is a smooth bundle map

$$\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$$

Definition 2.1.2 (Lagrangian Function)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian density. Define the Lagrangian function of $\mathcal{L}$ to be the unique function $L \in \mathcal{C}^\infty(J^1(E))$ such that

$$\mathcal{L} = L\mu_\eta$$

Recall: $j^1\phi : M \rightarrow J^1E$ is the jet prolongation of ϕ .

Definition 2.1.3 (Action Functional)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Define the action functional of $\mathcal{L}$ on Ω to be the map $S : \mathcal{C}^\infty(E) \rightarrow \mathbb{R}$ given by

$$S[\phi] = \int_\Omega \mathcal{L}(j^1\phi)$$

Definition 2.1.4 (Admissible Variations)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\Omega \subseteq M$ be compact. Define the set of admissible variations on Ω to be the set

$$\mathcal{C}_{\partial\Omega}^\infty(E|_\Omega) = \{\delta\phi \in \mathcal{C}^\infty(E|_\Omega) \mid \delta\phi|_{\partial\Omega} = 0\}$$

Definition 2.1.5 (First Variation)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Let $\phi \in \mathcal{C}^\infty(E)$. Let $\delta\phi \in \mathcal{C}_{\partial\Omega}^\infty(E|_\Omega)$ be an admissible variation. Define the first variation of ϕ along $\delta\phi$ by

$$\delta S[\phi, \delta\phi] = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S[\phi + \varepsilon\delta\phi] \in \mathbb{R}$$

Definition 2.1.6 (The Euler-Lagrange Operator)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Define the Euler-Lagrange operator $\text{EL} : \mathcal{C}^\infty(E) \rightarrow \mathcal{C}^\infty(\text{Hom}(E, \text{Den}(M)))$ as follows. Let $(U, x^1, \dots, x^4)$ be a chart of M . Let $(p^{-1}(U), x^i, y^j)$ be a local trivialization chart of E . Let $(p^{-1}(U), x^i, y^j, y_i^j)$ be a local chart on $J^1(E)$ where $y_i^j \in \mathcal{C}^\infty(J^1(E))$ is such that $y_i^j(j_p^1\phi) = \left. \frac{\partial y^j \circ \phi}{\partial x^i} \right|_p$. Then locally at the chart, the operator $\text{EL}(\phi)_p : E_p \rightarrow \text{Den}_p(M)$ is given by

$$\text{EL}(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v) \left(\frac{\partial L}{\partial y^j} \circ j^1\phi - \sum_{i=1}^4 \frac{\partial}{\partial x^i} \left(\frac{\partial L}{\partial y_i^j} \circ j^1\phi \right) \right) \Big|_p \mu_\eta$$

where $\phi \in \mathcal{C}^\infty(E)$ and $v \in E_p$.

To explain:

- For $v \in E_p$, $y^j(v)$ picks out the j th coordinate of v according to our local chart.
- $\frac{\partial L}{\partial y^j}$ is a smooth function of $J^1(E)$. We may compose with $j^1\phi : M \rightarrow J^1(E)$ to obtain a smooth function from M to $\mathbb{R}$.
- Similarly, $\frac{\partial L}{\partial y^i} \circ j^1\phi$ is a smooth function from M to $\mathbb{R}$. Using the local chart of M , we may differentiate this function with respect to x^i to obtain another smooth function from M to $\mathbb{R}$.
- Notice that everything before the Minkowski density is indeed a real number. Hence $\text{EL}(\phi)_p(v)$ is indeed an element of $\text{Den}_p(M) \cong \mathbb{R}$.

Theorem 2.1.7 (Euler-Lagrange Theorem)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\phi \in \mathcal{C}^\infty(E)$. Then we have

$$\text{EL}(\phi) = 0$$

if and only if $\delta S[\phi, \delta\phi] = 0$ for all $\delta\phi \in \mathcal{C}_{\partial\Omega}^\infty(E|_\Omega)$.

2.2 Hamiltonian Density

Definition 2.2.1 (Conjugate Momentum Density)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let $(p^{-1}(U), x^i, y^j, y_i^j)$ be a local chart of $J^1(E)$. Define the conjugate momentum density $\pi_t : \mathcal{C}^\infty(E) \rightarrow \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t)))$ of $\mathcal{L}$ at t at the local chart by

$$\pi_t(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v) \left(\frac{\partial L}{\partial y_1^j} \circ j^1\phi \right) \Big|_p \mu_\eta|_{\Sigma_t}$$

Definition 2.2.2 (Hamiltonian Density)

Let $p : E \rightarrow M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let u_p be the corresponding inertial observer field. Define the Hamiltonian density of $\mathcal{L}$ to be the function $\mathcal{H}_t : \mathcal{C}^\infty(E|_{\Sigma_t}) \times \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t))) \rightarrow \mathcal{C}^\infty(\text{Den}(\Sigma_t))$ given by

$$\mathcal{H}_t(\phi)_p = (\pi_t(\phi)_p(j^1\phi(p)(u_p)) - L \circ j^1\phi(p)) \mu_\eta|_{\Sigma_t}$$

Definition 2.2.3 (The Hamilton Operator)

Theorem 2.2.4 (Hamilton's Equations)

2.3 Covariant Hamiltonian Formalism

2.4 Scalar Fields and Klein-Gordon Lagrangian

Definition 2.4.1 (Klein-Gordon Lagrangian Density)

Define the Klein-Gordon Lagrangian density to be the bundle map $\mathcal{L}_{\text{KD}} : J^1(M \times \mathbb{R}) \rightarrow \text{Den}(M)$

given at local chart $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ by

$$\mathcal{L} = \left(\frac{1}{2} \sum_{i=1}^4 \sum_{j=1}^4 \eta_{ij} y_i y_j - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

where (η_{ij}) is the matrix representation of η and $m \in \mathbb{R}$.

Lemma 2.4.2

Let $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\mathcal{L}_{\text{KD}} = \left(\frac{1}{2} ((y_1)^2 - (y_2)^2 - (y_3)^2 - (y_4)^2) - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

Lemma 2.4.3

Let $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\text{EL}(\phi)_p(v) = y(v) \left(-m^2 \phi(p) - \left(\frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2} \right) \right) \mu_\eta$$

Definition 2.4.4 (d'Alembertian)

Define the d'Alembertian operator $\square : \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$ by

$$\square \phi = \frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2}$$

Theorem 2.4.5 (Klein-Gordon Equations)

Let $\phi \in \mathcal{C}^\infty(M \times \mathbb{R})$. Then $\text{EL}(\phi) = 0$ if and only if

$$(\square + m^2)\phi = 0$$

Proposition 2.4.6 (Klein-Gordon Hamiltonian)

2.5 Noether's Theorem

Definition 2.5.1 (Continuous Symmetry)

Theorem 2.5.2 (Noether's Theorem)

3 Electromagnetism

Classical field theory is developed to accommodate discoveries in the field of electromagnetism. We demonstrate the power of this language.

The Observation of Charge

Certain objects, when rubbed together, exert forces on one another across empty space. A glass rod rubbed with silk attracts a rubber rod rubbed with fur. Two glass rods rubbed with silk repel each other. Two rubber rods repel each other.

This observation suggests that matter can acquire a property which causes it to exert and respond to forces. We call this property *electric charge*. The pattern of attraction and repulsion suggests that charge comes in two varieties. We label them positive and negative, with the convention that like charges repel and opposite charges attract.

A key further observation is that charge is *conserved*: in any isolated process, the total charge never changes. When a glass rod is rubbed with silk, charge is not created — it is transferred from one object to the other, leaving the total unchanged.

Coulomb's Law and the Electric Field

In 1785, Charles-Augustin de Coulomb measured the force between charged spheres using a torsion balance. He found that the force between two point charges q_1 and q_2 separated by a distance r is:

$$F = k_e \frac{q_1 q_2}{r^2}$$

where k_e is a constant of proportionality. The force is repulsive if the charges have the same sign, and attractive if they have opposite signs. This is Coulomb's law.

The *electric field* is the conceptual tool that makes sense of how this force is transmitted. Rather than saying that charge q_1 acts directly on charge q_2 at a distance, we say that q_1 modifies the space around it, producing an electric field $\mathbb{E}$. The charge q_2 then responds to the field it finds itself in, not directly to q_1 . The field is defined operationally as the force per unit charge exerted on a small positive test charge:

$$E(x) = \lim_{q \rightarrow 0} \frac{F(x)}{q}$$

The limit ensures we do not disturb the field we are trying to measure. The electric field is physically real: it carries energy and propagates at finite speed, so distant charges cannot influence each other instantaneously.

Magnetism and Faraday's Discovery

Magnetic phenomena were known long before electromagnetism was unified. Lodestones attract iron. Compass needles align with the Earth's field. For centuries, electricity and magnetism appeared to be entirely separate phenomena.

In 1820, Hans Christian Orsted discovered that a current-carrying wire deflects a compass needle placed nearby. A moving charge produces a *magnetic field* B in the space around it, just as a static charge produces an electric field. Andre-Marie Ampere then showed quantitatively that parallel currents attract, and anti-parallel currents repel.

In 1831, Michael Faraday made the reciprocal discovery: a changing magnetic field drives a current in a nearby wire. He described this in terms of *lines of force* threading through a loop. When the number of these lines changes — whether by moving a magnet, changing a current, or moving the loop itself — a

voltage is induced in the loop. This is electromagnetic induction.

Faraday also observed that magnetic field lines always form closed loops. No matter how a magnet is cut or arranged, one never finds an isolated magnetic “charge” (a *monopole*) from which field lines emanate. This asymmetry between electricity and magnetism — electric charges exist, magnetic charges do not — is a fundamental observational fact.

The Unification

The deepest insight of the theory, clarified by Einstein’s special relativity, is that electric and magnetic fields are not truly separate objects. They are two facets of a single *electromagnetic field*, seen from different reference frames.

An observer watching a stationary charge sees only an electric field. A second observer moving relative to the first sees the same charge in motion — a current — and therefore sees both an electric and a magnetic field. The same physical situation produces different fields depending on the observer’s state of motion. This is not a paradox; it means the split between E and B is frame-dependent, while the underlying electromagnetic field is the invariant object.

Maxwell’s Equations

Experiment 1: Coulomb’s Law: The electric field produced by a point charge q at the origin is given by the map $E : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ where

$$E(x) = \frac{q}{4\pi\epsilon_0} \frac{x}{|x|^2}$$

for some constant $\epsilon_0 \in \mathbb{R}$.

Axiom 1: Gauss’s Law: Denote ρ by the charge per unit volume of $V \subseteq \mathbb{R}^3$. From the above we can deduce

$$\int_S E \cdot dS = \frac{1}{\epsilon_0} \int_V \rho dV$$

, which equivalently says that

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

Experiment 2:

$$\int_S B \cdot dS = 0$$

for a closed surface S .

Axiom 2: Gauss’s Law:

$$\nabla B = 0$$

Experiment 3: Faraday observed

$$\int_C E \cdot dl = -\frac{d}{dt} \int_\Sigma B \cdot dS$$

where C bounds the surface Σ .

Axiom 3: Faraday’s Law:

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Experiment 4: Ampere observed???

Axiom 4: Ampere Maxwell Law:

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

Theorem: the above implies conservation of charge.

Lorentz force law:

$$F = q(E + v \times B)$$

Something called the field potential F packs axiom 2 and 3 into a convenient package.

Notice that all the above are not time dependent. Einstein: when talking about how these fields evolve in time, the split between E and B is frame dependent, but the electromagnetic field is frame independent. (what this means: E and B come from F , F has a frame independent description, and we recover E and B by choosing a frame)

3.1 The Potential and Deriving the Classical Maxwell Equations

Definition 3.1.1 (The Four Potential)

A four potential is a smooth section $A \in \Omega^1(M)$.

We can reverse the process: the four potential gives the sourceless maxwell equations (2) and (3). Only if we assume conservation of charge, we can imply (1) and (4).

Definition 3.1.2 (The Field Tensor)

Let $A \in \Omega^1(M)$ be a four potential. Define the field tensor of A by

$$F = dA \in \Omega^2(M)$$

Definition 3.1.3 (Electric and Magnetic Fields)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field.

- Define the electric field of A to be

$$E = \iota_u F \in \Omega^1(M)$$

Explicitly, we have $E_p(v) = F(u, v)$.

- Define the magnetic field of A to be

$$B = \iota_u * F \in \Omega^1(M)$$

Explicitly, we have $B_p(v) = (*F)(u, v)$

Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding foliation (recall that they are the slices of M along a fixed time t). Then E and B restrict to each Σ_t which are the familiar electric and magnetic fields in classical electromagnetism. The restriction does not loose information because the time component is 0.

Definition 3.1.4 (Classical Electric and Magnetic Fields)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion.

- Define the electric field at time t to be

$$E_t = \iota_t^* E \in \Omega^1(\Sigma_t)$$

- Define the magnetic field at time t to be

$$B_t = \iota_t^* B \in \Omega^1(\Sigma_t)$$

Definition 3.1.5 (The Four Current of a Potential)

Let $A \in \Omega^1(M)$ be a four potential. Define the four current to be

$$J = *^{-1} d * F \in \Omega^1(M)$$

Definition 3.1.6 (Charge Density)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Define the charge density of A by

$$\rho = \iota_u J \in \mathcal{C}^\infty(M)$$

Theorem 3.1.7 (Classical Maxwell's Equations)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion map. Then the following are true.

- Gauss's law for electricity: $d_{\Sigma_t} *_{\Sigma_t} E_t = \iota_t^* * J$.
- Gauss's law for magnetism: $d_{\Sigma_t} B_t = 0$.
- Faraday's Law: $d_{\Sigma_t} E_t + \frac{\partial B_t}{\partial t} = 0$.
- Ampere-Maxwell Law: $d_{\Sigma_t} *_{\Sigma_t} B_t - \frac{\partial E_t}{\partial t} = \iota_t^* J_{\Sigma_t}$.

3.2 The Maxwell Lagrangian

Definition 3.2.1 (The Maxwell Lagrangian)

Define the Maxwell Lagrangian to be the map $\mathcal{L}_{\text{ML}} : J^1(T^*M) \rightarrow \text{Den}(M)$ given by

$$\mathcal{L}(j^1 A)_p = -\frac{1}{4}(dA)_p \wedge *(dA)_p$$

Proposition 3.2.2

The Euler-Lagrange operator of the Maxwell Lagrangian is given by

$$\text{EL}_{\text{ML}}(A) = d * dA - *J$$

3.3 Computing the Potential From a Current

Starting with A , we defined all the other electromagnetic quantities. Physically, we start directly with the electromagnetic field and deduce all the other quantities. On the other hand, we can start with J , and ask what the other quantities are. This is more natural because physically, starting with J is the same as specifying the matter distribution (where the charges are and how they move).

Postulate: Conservation of charge: $d * J = 0$.

Definition 3.3.1 (The Four Current)

A four current is a section $J \in \Omega^1(M)$ such that $d * J = 0$.

If we started with a four potential A , recall that J is defined by the formula $d * J = d * dA$.

Proposition 3.3.2

Let $J \in \Omega^1(M)$ be a four current. Let $A \in \Omega^1(M)$. Then $d * dA = d * J$ if and only if $\text{EL}_{\text{ML}}(A) = 0$.

The problem is that we cannot just take this as the definition of the associated potential, because A is not unique. Let $f \in \mathcal{C}^\infty(M)$. Then notice that if A is a solution, then $A + df$ is also a solution. We choose a common potential as definition (so other definitions can also possibly work, but we use the following one as the standard one).

Definition 3.3.3 (Gauge Fixing)

This gives some boundary conditions so that the pde $d * J = d * dA$ is well posed.

The solution to the pde is given as follows:

Definition 3.3.4 (The Retarded Potential)

Let $J = \sum_{i=1}^4 J_i dx^i \in \Omega^1(M)$ be a four current. Define the retarded potential of J to be the one form $A = \sum_{i=1}^4 A_i dx^i$ whose components are given by

$$A_i(p_1, \dots, p_4) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{J_i(p_1 - \sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}, x_2, x_3, x_4)}{\sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}} dx_2 dx_3 dx_4$$

4 General Relativity

Equivalence Principle: Dynamics Postulate:

Minimal-coupling principle.

Definition 4.0.1 (Spacetime)

Spacetime is an n -dimensional orientable pseudo-Riemannian manifold (M, g) where g has signature $(n - 1, 1)$.

Definition 4.0.2 (Einstein-Hilbert Action)

Whats next: Classical field theory (which includes relativistic field theory), continuum mechanics (which includes relativistic quantum mechanics).

5 The Logical Chain of Classical Field Theory and Gauge Theory

Step 1: The Field

We observe a physical quantity distributed over spacetime — a temperature, a velocity, an electromagnetic strength. Mathematically this is a fiber bundle $\pi : E \rightarrow M$, where M is spacetime and the fiber E_x is the space of possible values of the quantity at x . A **section** $\phi \in \Gamma(E)$ is one complete specification of the quantity at every point — a possible physical configuration.

Step 2: The Lagrangian as a Choice of Physics

A Lagrangian density $\Lambda \in \Omega^n(J^k E)$ is a choice of action functional

$$S[\phi] = \int_{\Omega} (j^k \phi)^* \Lambda$$

Different Lagrangians encode different physical theories. The Euler-Lagrange equations — the equations of motion — are derived from Λ by the variational principle. The physical sections are the critical points of S .

Step 3: Observing a Symmetry

You compute the Euler-Lagrange equations and notice that the Lagrangian Λ is invariant under some subgroup of automorphisms of E . An **automorphism** of $\pi : E \rightarrow M$ is a fiber-preserving diffeomorphism $\Phi : E \rightarrow E$. Such an automorphism acts on sections by

$$\phi \mapsto \Phi \circ \phi$$

and the Lagrangian being invariant means

$$S[\Phi \circ \phi] = S[\phi] \quad \text{for all } \phi \in \Gamma(E)$$

At this stage, the symmetry group acts the same way at every point — it is **global**. You observe this as an empirical fact about your Lagrangian.

Symmetries of the action induces certain laws of physics, such as conservation of energy, momentum, charge etc. This is Noether's theorem.

Another note: Rather than starting with a Lagrangian and observe the symmetries of its action, we may also start with a set of symmetries that we would like to enforce, apply it to all differential forms and retain only the invariants, and see what Lagrangian you would get.

Step 4: The Absolute vs Relative Observation

The global symmetry exists because the field takes values in a space that has **no preferred origin or frame** — the fiber E_x has a group of self-maps (rotations, phase shifts, color rotations) that are physically unobservable. Different observers, or different conventions, would describe the same physical situation using sections related by this group action. The global symmetry is an artifact of having made consistent conventional choices everywhere at once. We need not restrict ourselves to consider symmetries of the entire space, and this is supported by physics:

Argument 1: Special Relativity

Special relativity says there is no notion of simultaneity — two spacelike separated points have no canonical notion of “the same moment.” A global transformation is therefore not a physically coherent operation in a relativistic theory. If you believe in special relativity, you are forced to allow your symmetry to act differently at different points. The global symmetry is then a special case of the local one, not the fundamental object.

Argument 2: The Gauge Principle

The difference induced by the symmetry of the action is not observable. The phase is therefore a **convention** — a choice made by an observer at x to describe the field. If the phase is a convention, then different observers at different points of spacetime should be free to make different conventional choices independently. There is no physical mechanism that would force their conventions to agree globally. The true symmetry group of the theory is therefore not the global group G but the local group $\text{Map}(M, G) = \text{Map}(M, G)$ of all smooth G -valued functions on spacetime.

This is the **gauge principle**: any redundancy in the description of a physical system should be a local redundancy. It is a postulate, not a theorem. Its justification is that it works — every time it is applied, it produces a physically correct and predictive theory.

Step 5: Promoting to Local Symmetry

This forces you to demand invariance not just under global G -transformations

$$\phi(x) \mapsto g \cdot \phi(x), \quad g \in G \text{ constant}$$

but under local G -transformations

$$\phi(x) \mapsto g(x) \cdot \phi(x), \quad g : M \rightarrow G \text{ smooth}$$

The Lagrangian, built from ordinary derivatives $\partial_\mu \phi$, immediately fails to be locally invariant. The reason is fundamental: $\partial_\mu \phi$ compares field values at nearby points x and $x + dx$, which under local symmetry live in fibers with **independent conventions**. Subtracting them is meaningless — it is comparing quantities measured in different units.

Step 6: The Mathematics Is Now Forced

To make sense of differentiation under local symmetry, we need a way to **compare vectors in nearby fibers** compatibly with the G -structure. But this is precisely the definition of a **connection**. Where does this connection live? The answer is that the principal bundle is not introduced from outside — it is **already contained inside the vector bundle** E .

The fibers E_x are vector spaces, but the isomorphism $E_x \cong \mathbb{R}^r$ is not canonical — it requires a choice of basis. A **frame** at x is an ordered basis for E_x . The set of all frames at x carries a natural free transitive right action of $GL(r, \mathbb{R})$, making it a $GL(r, \mathbb{R})$ -torsor. Taking the disjoint union over all $x \in M$ produces the **frame bundle**

$$P := \bigsqcup_{x \in M} \{\text{ordered bases for } E_x\}$$

a principal $GL(r, \mathbb{R})$ -bundle canonically constructed from E .

Now, the observation of a global G -symmetry in Step 3 is precisely the observation that the physically relevant frames are not arbitrary frames, but only those related by G -transformations. The structure group **reduces** from $GL(r, \mathbb{R})$ to G , producing a principal G -bundle

$$P_G \subset P$$

canonically determined by the symmetry. The original vector bundle is recovered as the associated bundle $E = P_G \times_G V$.

The connection demanded by local symmetry lives on P_G , because P_G is the bundle whose fibers are G -torsors — they encode exactly the gauge freedom at each point. A connection on P_G transports G -frames between nearby fibers, and therefore transports everything associated to those frames, including sections of $E = P_G \times_G V$.

Step 7: What Follows From The Maths

Since the principal bundle is just a collection of frames at each fiber. A local section of the principal bundle is a choice of local coordinates. The mathematical fact that non-trivial principal bundles have

no global sections can be interpreted as the fact that there is generally no globally consistent choice of coordinates because the symmetries now act locally.

A connection in the principal bundle gives a covariant derivative in the associated bundle. This gives a way to compare phases across different time space.